



OPEN Blockchain-based energy consumption approaches in IoT

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Blockchain is a decentralized electronic account book that is secure, open, and impervious to manipulation, and records transactions across numerous computers. Internet of Things (IoT) deployed sensors are limited in bandwidth, processing power, and battery life. Energy efficiency is a vital factor and a decisive matter for energy-bound IoT-based networks. This paper provides an idea of the working principle of blockchain technology in IoT-enabled energy systems. It investigates the fundamentals of blockchain technology, clarifying its decentralized nature, cryptographic mechanisms, and consensus algorithms that ensure data immutability and transparency. It explores how blockchain technology can be integrated with various IoT fields for energy efficiency. The paper investigates the blockchain-based energy consumption approaches in IoT qualitatively, and quantitatively with future research directions and challenges. It is seen that the hybrid blockchain model is suitable for energy efficiency in IoT.

Keywords AI, Blockchain, Energy, IoT, UAV

The IoT is an increasingly popular area of study due to advances in wireless communications and sensor technologies. This development has an impact on a wide range of industries and sectors, including healthcare, different business domain names the smart sector, etc. Because so many various kinds of heterogeneous devices are interconnected in these fields, safeguarding is a major concern¹. A vast number of tiny, battery-operated sensors that gather and transmit environmental data make up IoT. These days, a range of IoT attacks hazard its adoption. The majority of attacks that are currently in use include denial of service (DoS), spyware, malware, phishing, and zero-day attacks². The attacks could be passive or active, internal or external. A minimum of one of these incidents affects every layer of the IoT. DoS attacks affect all IoT tiers, avoidance of users from accessing the network.

Blockchain is seen as an indwelling security technology that offers confidence, safety, confidentiality, and anonymity to IoT networks^{3,4}. Numerous issues with conventional distributed database systems are addressed by it. Data generated by IoT devices has security features thanks to the blockchain' unbreakable structure⁵. The process of mining creates the block, which contains the data that are contained within. The process of mining the most popular consensus method, known as proof of work, requires a lot of energy. The process of repeatedly performing a cryptographic hashing operation until the hash of a certain amount of initial zeroes appears is one of the primary causes of the massive energy consumption⁶. Researchers are now very interested in the quantity of electricity used for proof of work (POW)-based blockchain mining. Blockchain hash computation uses a lot of energy i.e., 205.322 terahertz per second is the average hash rate at the moment⁷. Owing to the high number of hashes produced every second, hash generation consumes a tremendous amount of energy.

When blockchain is integrated with IoT networks, accurate information provision is guaranteed. By using a blockchain approach, services can be operated and distributed, eliminating the need for an outside entity to act as a middleman⁸. Since the blockchain network excludes several non-colluding parties, it is the most efficient way of fostering trust within the IoT. Because most IoT nodes are low-powered, they must delegate the computation to other devices. Compute-intensive tasks are typically offloaded to nearby edge devices, fog machines, cloud devices, multiparty providers of services, etc.⁹. Computing power is delivered by computing at the edge, which sits right next to the terminal devices. By providing a high level of latency, computing power may be made available to technologies that require a large number of computations¹⁰. When offloading, latency as well as energy consumption, network capacity, and allocation of resources are the primary factors to take into account^{11,12}.

The goal of creating opportunistic, energy-efficient clustering algorithms is to solve the problems associated with energy consumption in networks with limited resources, such as IoT communications. Given these sensors' limited energy resources, extending the network's lifetime while preserving its ability to function is a crucial

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concern. Opportunistic environmentally friendly blockchain techniques use the network's inherent properties, communication habits, and node heterogeneity to their advantage to minimize energy consumption.

IoT describes a network composed of real objects or things with software, sensors, and connectivity capabilities that allow them to gather, swap, and act on data. These physical objects can be daily items like home appliances, wearable devices, or industrial equipment, all enhanced with internet connectivity to facilitate communication between devices and with centralized systems³. Blockchain is a revolutionary technological advance that functions to be a decentralized and distributed accounting database. It treats a network of nodes or computers where each node maintains an identical copy of a continuously growing list of records, known as blocks. These blocks are linked together using cryptographic principles to form a chain, hence the name "blockchain"⁴. Such structure assures that when data is entered in a block, it is extremely difficult to alter retroactively without varying entirely collateral blocks, thereby supplying a rich level of immutability and security⁵. Artificial intelligence (AI) is the simulation of cognitive ability in machines that have been designed to think as human beings and perform their activities. It covers a wide range of innovations and methods, including natural language processing, computer vision, machine learning, and robotics. Unmanned Aerial Vehicles (UAVs), also known as drones, are airplanes that fly without a human pilot on board. They can be controlled remotely by an individual or automatically by onboard computers. UAVs are outfitted with a variety of sensors, cameras, and, in some cases, weapons for military use. They are used for a variety of applications, including surveillance, reconnaissance, monitoring, goods transport, science research, and even business operations such as filming and package delivery¹³.

Peer-to-peer energy switching and increased transaction transparency are two ways that blockchain-based energy consumption strategies in the IoT enable decentralized energy management. By automating energy¹⁴ transactions with smart contracts, these systems modify real-time supervising of energy production and expenditure while maintaining data integrity through immutable blockchain records. This innovation solves inefficiencies in conventional energy markets and maximizes the use of renewable energy sources, enabling users to directly trade excess energy with one another. In the end, this helps create a more alive and sustainable energy ecosystem¹⁵. Concerns regarding the environmental effects¹⁶ of blockchain-based energy systems have been highlighted by the energy-intensive process of validating and appending new transactions to the blockchain.

Motivation

The main objective is to address how blockchain technology is used to minimize energy consumption in IoT and also provide the current research challenges for it. This article gives an idea of the working principle of blockchain technology in IoT-enabled energy systems. It investigates the fundamentals of blockchain technology, clarifying its decentralized nature, cryptographic mechanisms, and consensus algorithms that ensure data immutability and transparency. It exposes how blockchain technology can be integrated with various IoT fields for energy efficiency. The motive of this survey is to stimulate the researchers to utilize the blockchain-based concepts and sort out the issues that need further research. The main motivation of the article is to grasp the knowledge about energy consumption in blockchain-based IoT for the researchers.

Contribution

The main contributions of the article are as follows:

- Multiple blockchain-based IoT techniques are critically analyzed in terms of qualitatively, and quantitatively with several well-defined parameters like energy, and blockchain.
- This article also investigates the advantages and drawbacks of blockchain-based energy consumption approaches in IoT.
- It is observed that the hybrid blockchain model is effective for energy in IoT.

Paper organization

Section "[Blockchain technology](#)" summarizes blockchain technology. Section "[Integration of IoT with blockchain technology](#)" discussed the integration of IoT with blockchain technology. Section "[Related work](#)" addresses state-of-the-art blockchain-based energy consumption approaches in IoT. Section "[Challenges of blockchain-based energy consumption approaches in IoT](#)" illustrates the challenges of blockchain-based energy consumption approaches in IoT. Section "[Future research direction](#)" demonstrates future research direction. Section "[Insight thought from this article](#)" describes insight thought from this article. Lastly, Section "[Conclusion](#)" concludes the article.

Blockchain technology

A computer is linked to the blockchain and the computer is called a node. Node uses the client to establish a connection with the blockchain. The client contributes to the propagation of transactions on the blockchain and their validation. A copy of all blockchain data is downloaded into the network when a computer establishes a connection to the blockchain, allowing the node to stay up to date with the most recent block of data. Miners are the nodes that have links to the blockchain and that, in exchange for rewards, assist in the successful completion of transactions.

Working principle of blockchain technology

Bitcoin has become one of the more popular applications of blockchain. Bitcoin, a cryptocurrency, is a means of online asset exchange. Bitcoin's cryptographic proof enables two parties to carry out transactions over the World Wide Web without the need for third parties. Every transaction is protected by the use of a digital signature. Table 1 describes the advantages of blockchain technology in real life. Figure 1 shows the flowchart of blockchain to understand how they work in blockchain environment.

Key points	Description
Time-saving	The process is quicker and less expensive because settlements do not require central authority verification
Cost-saving	There are multiple ways in which a blockchain network lowers costs and a few middlemen. Transaction efforts are cut down as ledgers are shared among all the participants
Increased security	Blockchain data is shared between millions of participants. So, it is impenetrable to tampering
Collaboration	It ignores the need for third-party negotiation as a result permits straight communication among the parties
Reliability	The party's identity is authenticated and certified. Identical records are removed

Table 1. Advantages of blockchain technology.

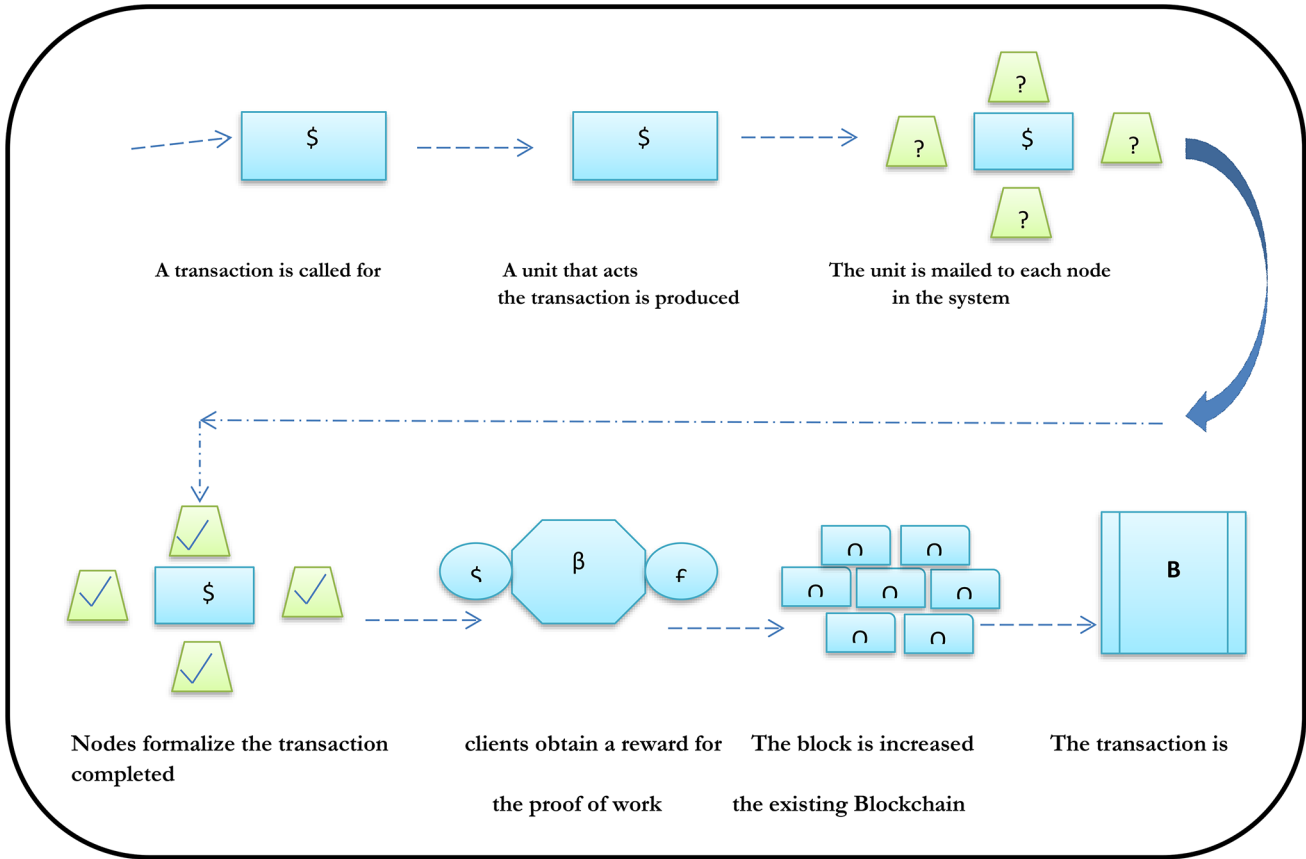


Fig. 1. Flowchart of blockchain's work.

Integration of IoT with blockchain technology

The consolidation of IoT with blockchain technology represents a transformative synergy that revolutionizes data management and security in the digital age. By marrying the capabilities of IoT devices with the decentralized and immutable ledger of blockchain, this integration ensures unprecedented levels of trust, transparency, and efficiency in various applications like in the field of financial transactions, health care, defense systems, transport control, e-commerce systems etc. IoT devices, ranging from sensors to smart appliances, collect vast amounts of data about their surroundings, which is then securely encrypted and transmitted over networks. Instead of relying on centralized servers, this data is stored on a blockchain, where each transaction is cryptographically linked and distributed across a network of nodes, ensuring tamper-proof records. Smart contracts further automate processes, enabling seamless interactions between IoT devices while ensuring compliance with predefined rules. This integration not only enhances data integrity and privacy but also facilitates real-time tracking, transparency, and traceability across supply chains, healthcare systems, energy grids, and beyond. Overall, the integration of IoT with blockchain technology empowers industries with a robust foundation for innovation, collaboration, and sustainable growth in the digital era. Here's a simplified description of how this integration typically works as follows:

- 1. Data gathering by IoT nodes:

IoT devices, such as nodes or sensors, actuators, and other smart gadgets, etc. collect and transmit data about their environment or the objects they interact with.

2. Data encryption and transmission:

Before transmitting data, IoT devices encrypt it to ensure its security and privacy. Encrypted data is then sent over a network, such as the internet or a local network, to centralized servers or other IoT devices.

3. Data storage on the blockchain:

Instead of storing data on centralized servers, IoT data is stored on a blockchain, which is a decentralized and immutable ledger.

4. Blockchain consensus mechanism:

Transactions on the blockchain are validated and confirmed through a consensus execution, like proof of work (PoW), proof of stake (PoS), etc. Once formalized, transactions are increased on the blockchain, making them tamper-proof and transparent.

5. Intelligent contracts for automation:

Smart contracts, or autonomous agreements with predetermined conditions created using code, can be executed on the blockchain. IoT devices can work with smart contracts to perform automated tasks like sharing data, money transfers, managing a supply chain, and more.

6. Data traceability and transparency:

Because blockchain is immutable and transparent, every data transaction recorded on the blockchain can be described and endorse to its beginning.

7. Enhanced security and decentralization:

Blockchain's decentralized and cryptographic features enhance the security and privacy of IoT data. Since data is stored across multiple nodes in the network, it is less vulnerable to individual points of loss and hacking attempts. Additionally, encryption techniques and private keys ensure that only authorized parties can access and interact with the data. By leveraging blockchain technology, IoT networks can become more decentralized, resilient, and scalable.

Related work

The selected review articles covered a variety of features, such as the fundamental ideas of different blockchain-based energy consumption approaches in IoT by applying various parameters i.e. clustering mechanism, energy consumption, number of sensors, tools, use of methodology, etc., and simulation methods. Similar types of existing models are helped to analyze in the literature review for understanding blockchain-based energy efficiency in IoT.

The consensus method of blockchain for edge computing in IoT networks is described by the authors¹⁷ as being to a greater extent power-efficient, requiring deficient storage and interval time. When similar forms of elements are found in the boundary web, the limitations of such acts become apparent. Future iterations of such an act could involve applying this experiment to actual entities in the real world. To improve this strategy across different blockchain environments, such work may also be that applies to the consortium blockchain. The authors¹⁸ described an energy-efficient software-defined networking (SDN) control architecture for IoT networks with blockchain-based security, which uses a bunch structure and a routing protocol. To remove Proof-of-Work (POW) in P2P (peer-to-peer) communication between IoT devices and SDN controls, the architecture employs private as well as public blockchain technologies. Furthermore, it used an effective authentication technique with circularized belief, constructing the blockchain suitable for IoT devices with a restricted budget. It was demonstrated that this architecture outperforms traditional blockchain, but it doesn't seem practical.

Hybrid blockchain technology for IoT was discussed by the authors¹⁹ about a reward system intended to motivate blockchain agents to contribute to the H-chain while simultaneously taking energy consumption into account. The work presented the framework in a scenario that secured a wearable device connected through an SDN, even though the work tends to apply to IoT applications. First, the evaluation results offered a way to equilibrium the public and private portions of any hybrid-blockchain (H-chain) deployment based on various factors such as verification complexity, computation capability, and network conditions. The results of the simulation show that, even with the hybrid consensus mechanism's high energy consumption, the advantage design can motivate blockchain agents to make contributions to the H-chain. As a result, the mentioned H-chain can achieve ideal social welfare despite its subpar performance.

To categorize both the thoughts and proficient elements of eco-friendly energy methods established on the use of blockchain in mobile gathering recognizing concerning technical details, utilized techniques, evaluation variables, and evaluation surrounds, the authors²⁰ disclosed that blockchain technology to support energy-aware mobile gather sensing approaches in the IoT regarding classify but cannot reach a satisfactory solution. In their work, the authors²¹ presented a blockchain-based energy trading scheme that is both secure and efficient. To significantly increase system availability, a credibility-based fairness-proof process is designed to adjust the comparatively low processing power of the underlying IoT devices.

The authors²² provided evidence that a blockchain-based bright result is available for demand-side administration of energy in residential buildings within a region. This solution improved peaks to the average ratios (PAR) for electrical load, lowered energy consumption, and improved occupants' thermal comfort through appliance, lighting, and heating system modeling. A transient mathematical physical representation had been developed for the neighborhood's real-time monitoring of temperature and power. The authors²³ showed that there is still a substantial gap in benchmarking historical data trading procedures in IoT environments because of the lack of knowledge regarding the effectiveness of communication between sellers and buyers in modeling and analysis of information trading on blockchain-based the market in IoT Networks. Driven by this knowledge void, the authors presented a model for massive environmental sensing via narrowband Internet-of-things (NB-IoT)-based distributed ledger technologies (DLT) data trading.

The authors¹³ showed how an energy-efficient gathering device for unmanned aerial vehicles (UAVs) that is enabled by blockchain technology helps the IoT detect UAV swarms and creates distributed databases based on blockchain technology to fend off intruder UAVs. They created an adaptive linear forecasting algorithm to lower energy usage. By using this algorithm, in-network transmissions are significantly decreased as IoT devices submit prediction models rather than original data. To improve the dependability of smart city networks for sensitive data, scientists have proposed the blockchain mechanism²⁴. Effective resource scheduling is one of the main problems and difficulties with sensor networks in smart cities, as it prolongs the network life. With its mobile nodes, this model is incredibly robust.

According to the author²⁵, artificial intelligence (AI) is designed to forecast load profiles and energy consumption in addition to scheduling resources to guarantee dependable performance and efficient use of energy resources. Massive amounts of data are needed to train AI models. AI performance is determined by the functions and relationships that can be found through the use of big data systems and information mining. The authors²⁶ bemoaned the lack of disclosure and tamper-resistance in the retention and retrieval of data in IoT networks caused by a blockchain-based strategy for IoT systems. An assessment of an established prototype showed that the suggested approach holds promise for providing a cohesive framework for the security of IoT data. IoT-based smart cities are made possible by blockchain technology, according to the authors²⁷. Consensus algorithms, blockchain platforms, and component technologies were used to illustrate the development of blockchain technology. They discovered the open research challenges, their main causes, and some potential fixes, but they weren't sufficient. According to the authors²⁸, monitoring real-time data concerning electricity system networks (ESN), AI, IoT, and blockchain provides peers with automated services that improve sustainability, stability, security, and availability.

The authors²⁹ investigated how blockchain technology and IoT sensors can be used together to create sustainable, futuristic energy solutions by efficiently monitoring both the electrical and physical parameters. Home appliances' daily and weekly electricity consumption can be assessed, and this solution aids in figuring out consumption rates and ways to lower energy expenses based on real-time data. The fundamentals of both technologies, as well as the architecture, protocols, and operation of the blockchain-based Internet of Things (BLoT) and a few BLoT applications that may be built on top of it, were covered in³⁰. A comparison was made, but no conclusive results were obtained. The effective lightweight integrated blockchain (ELIB) design was presented in³¹ to address IoT requirements. As a key example, the model was implemented within a smart home setting to confirm its suitability for a range of IoT scenarios. A smart home's resource-constrained resources benefit from a centrally located manager that creates shared keys for data transmission and handles all inbound and outbound requests. The results of the experiment showed that, under certain evaluation parameters, the ELIB performs at its peak, though not always.

The authors³² described a brand-new wirelessly powered edge intelligence (WPEG) structure that used energy harvesting (EH) techniques to produce stable, reliable, and sustainable edge intelligence. To safeguard the peer-to-peer energy and information sharing within the framework, the authors constructed a permissioned edge blockchain. Then explore the WPEG framework and develop the best edge learning path to maximal intelligence at the edge efficiency. Their simulation results demonstrated that, in comparison to traditional schemes, their incentive methods could maximize the utility of both parties. The optimal learning design, on the other hand, could achieve the highest possible learning efficiency, but at a high cost. According to the authors³³, this allowed for the dynamic creation of user-defined services without the need for network provider entities or intermediary services to authenticate or deliver composite services. Reinforcement learning is used in the composition process to create credible and safe composition paths. When participants solve intricate composition processes, cloud and fog things reward them. While defining a node's awareness and inherent characteristics helped determine its willingness to cooperate, it did not provide any assurance regarding the privacy of data.

The authors³⁴ discussed data analytics for the identification of fake reviews using supervised learning and this was addressed by the rapid growth in digital advertising transactions, including sales and purchases. The suggested model was trained using fewer features. The authors³⁵ mentioned the vehicle-to-vehicle (V2V) wireless energy-sharing scheme using blockchain. Here, electric vehicles (EVs) have minimized carbon dioxide (CO₂) emissions in the environment. Optimal pricing maximizes profits for EVs through a linear equilibrium strategy. This model could be implemented using deep learning techniques for energy sharing. The authors³⁶ discussed multi-agent-based decentralized residential energy management applying deep reinforcement learning and the

blockchain framework secures and protects residential energy data, enabling decentralized energy management. The authors³⁷ explained reinforcement learning for a multi-agent-based residential energy management system whose goal is to decrease the use of energy in a multi-carrier system while also minimizing consumer discomfort. The authors³⁸ discussed blockchain-directed real-time dynamic patterns for energy management schemes where the remix integrated development environment (IDE) simulates and analyzes blockchain-based results using deployed smart contracts by taking trade efficiency and data memory cost.

The authors³⁹ proposed non-fungible tokens (NFTs) to represent unique assets and designed an architecture that addressed disputatious issues like transaction cost, time complexity, etc. The decentralized smart city of things (DSCoT) model was discussed for smart contracts with energy consumption and the authors discussed the requirement for more security services, and validation for energy efficiency in IoT⁴⁰. The security achievements were substantiated through a relative analysis, validating distinct features and improvements over alternatives in⁴¹. The proposed approach provided a smart city solution that ensures energy efficiency in IoT networks.

There are various blockchain-based IoT techniques have been analyzed and it is noticed that there is a needed blockchain-based hybrid model for energy efficiency¹⁹. This article finds some drawbacks in previous work regarding energy optimization in IoT. When comparing different blockchain-based IoT techniques, one must evaluate their features with several parameters i.e. clustering mechanism, energy consumption, tools and use of methodology, etc. It has been seen by critical analysis that most of the blockchain-based IoT approaches for energy worked with different clustering mechanisms¹⁷. Few approaches did with high energy consumption. There are some advantages to the energy efficiency consensus approach acting significantly faster. There are many principles used like intra-clustering and recovery of rechargeable batteries which help to minimize energy consumption. Data collection for UAVs¹³ is cost-effective due to inexpensive experimental equipment.

Blockchain-based IoT remedies take advantage of blockchain technology's decentralized and immutable nature to improve network security, transparency, and efficiency. Here's a real-time example to demonstrate this concept. Walmart utilizes blockchain-based IoT to track food products. Walmart can track the journey of goods from farm to store shelves by embedding sensors in food shipments and incorporating them with blockchain technology. This enables Walmart⁴² and its clientele to verify product authenticity, quality, and freshness, as well as determine and recall contaminated goods as needed. The implementation of blockchain guarantees that all data related to the food goods' journey is secure, open, and immutable.

In a smart grid (SG) setting, the authors⁴³ highlighted the major benefits of AI-powered electric vehicle charging. It lowers costs, increases grid stability, encourages the integration of rechargeable energy roots, and enhances client happiness in general. It emphasizes the challenge of AI-powered autonomous cars and the difficulty of integrating AI in SGs. The authors⁴⁴ demonstrated that storing and handling the data produced by smart meters (SMs), the secure energy trading system (SETS), a blockchain-based distributed energy trading system (ETS) framework, is suggested. The authors mentioned that in the context of the Internet of Electric Vehicles (IoEVs), the Bayesian game-based technique⁴⁵ is used to determine the best energy pricing. In this case, the best pricing approach that increases the utilities of the energy purchaser and vendor in the energy dealing process and enhances overall benefit is a linear equilibrium method. Dedicated smart contracts (SCs) are used to implement game-based pricing to provide security, dependability, and credibility. Table 2 provides a comparative analysis of various blockchain-based IoT techniques.

It is analyzed from various IoT techniques in Table 2 and finds that there is a needed hybrid model for energy efficiency in IoT. And also find many more drawbacks in previous work regarding energy optimization techniques in IoT. When comparing different blockchain-based IoT techniques, one must evaluate their features, advantages, drawbacks, and uses. Here is a comparison of a few popular blockchain-based IoT methods. So, the decision to employ a blockchain-based IoT approach is guided by a number of variables, including use case requirements, regulatory compliance, and privacy concerns requirements. Every technique has advantages and disadvantages that should be carefully considered by organizations according to their priorities and requirements. It is found that blockchain-based IoT techniques minimize energy consumption significantly. Table 3 shows a qualitative analysis of various IoT techniques.

Here, is a qualitative analysis of various IoT techniques in Table 3 and it clarifies that previous works have been covered with energy efficiency in IoT. And also found that much more weak points in previous work regarding energy optimization techniques in IoT. Qualitative analysis of various IoT approaches in blockchain-based energy optimization involves examining how different techniques leverage IoT devices and blockchain technology to enhance energy efficiency, optimize energy consumption, and facilitate energy trading. Since, integrating IoT devices with blockchain technology offers numerous opportunities for optimizing energy usage, enabling decentralized energy trading, enhancing grid resilience, and promoting renewable energy adoption. It is noticed that blockchain-based IoT techniques minimized energy consumption markedly. Table 4 provides a comparison of simulation analysis with various blockchain-based IoT approaches.

Hence, show the comparison of simulation analysis with various blockchain-based IoT approaches in Table 4 and come to the point that previous works have been covered by various tools, number of sensors, and corresponding output for energy efficiency in IoT. And also find that so many dimensions in previous work regarding energy optimization techniques in IoT.

Challenges of blockchain-based energy consumption approaches in IoT

The IoT is easily at risk from security threats due to its low technical requirements and ease of construction. As such, security is a key issue for highly secret applications like telemedicine and military and medical diagnostics. Self-adjusting and dynamic IoT devices ought to be able to dynamically adjust to changing scenarios and contexts. Carefully monitoring the setting up of IoT gadgets that are dynamic and self-adapting. Data is sent and received by the cluster heads (CHs) to the base station (BS) node. The attacker can insert malicious data into the sensor node as an alternative to attacking it directly. Moreover, the BS node is unable to ascertain

Ref	Domain	Methodology	Primary Contribution	Cluster form(yes/No)	Advantage	Disadvantage	Parameter	Outcome
17	IoT network	Heuristic algorithm	Advance in power expenditure	Yes	Significantly faster	This is not a surprising result	Sensors, cluster	Improved the energy consumption inside the clusters
18	Blockchain for IoT	Meta Heuristic algorithm	software-defined networking (SDN) controllers for IoT networks	Yes	Negligible energy consumption	Experimented on limited resources like battery energy, bandwidth, and computing power	cluster	Minimize energy consumption
19	Energy reducing	Meta Heuristic algorithm	proposed a reward plan to incentivize the blockchain agents	Yes	stabilize independently	This is not a surprising result	Energy consumption on Circuit	Selection of cluster head was high accuracy
20	cellphone crowd sensing	Heuristic algorithm	Separate principles and expert aspects of eco-friendly energy schemes	Yes	Least energy consumption	Unavailability of data	Energy distribution	Minimize energy consumption
21	Industrial Internet of Things (IIoT) enable energy internet	Meta Heuristic algorithm	It defends the secrecy and achieve a balance of power supply	Yes	Least energy consumption	Low performance and Scalability	cluster	Minimize energy consumption
22	Energy used in a wireless network	Meta Heuristic algorithm	Maintaining the connectivity between the clusters and extending the lifespan of CHs nearer the base station are the goals of EDDUCA	Yes	Negligible energy consumption	It depends on the node in which the highest weight in each cluster is chosen to be the CH	spatial clustering approach	best (also strongest) peer cluster distribution
23	Energy reducing	Heuristic algorithm	Energy load balancing and the election of CHs	Yes	Least energy consumption	Allocating a maximum limit of CMs for a CH according to its remaining power and count of neighbors	Latency, energy	Find latency and energy consumption
13	Lifetime of clusters	Meta Heuristic algorithm	Energy Efficient intra-clustering	Yes	Good network performance	Limited increases in intra-clustering efficiency	energy	Improve the security and efficiency of data collection
24	Cluster formation	Hybrid algorithm	Energy reducing	Yes	Good network performance	The value of the cost function is updated regularly, not randomly	packet delivery ratio, throughput	Increases the efficiency in terms of the number of live nodes
25	Blockchain in smart energy	Meta Heuristic algorithm	Energy reducing	No	Negligible energy consumption	Low performance and Scalability	Blockchain, energy	Efficient and seamless integration of artificial intelligence
14	Cyber-Physical Systems	Digitization and Authentication in IoT	To generate an authentication access code based on extended non-fungible tokens (NFTs)	No	decentralized blockchain results	Low activity for the dynamic nature of smart city	execution cost	Used to authorize access
15	Authentication of IoT pluses in smart cities	non-fungible tokens (NFTs)	smart city solution for cyber-physical systems	No	extensive version of NFTs as unique	The authors did not use any outside dataset for simulation	Time length	To make a digital authority of smart devices such as IoT

Table 2. Comparative analysis of various blockchain-based IoT techniques.

whether the data that has been received is trustworthy or not. The IoT may experience some silent attacks, which are very dangerous since they introduce fake data in routing. Cluster congestion or bottlenecks are caused by this unnecessary redundant data. The packet dropping and delay cause the communication channel to become unreliable and inefficient overall. Blockchain-based energy consumption strategies in IoT encounter several warning challenges. To begin, scalability continues to be a critical issue, as blockchain systems have trouble handling the enormous number of operations produced by IoT gadgets in real time⁴⁶. Due to shortages of resources, low-power IoT-enabled digital belongings for infrastructure in smart cities necessitate blockchain solutions. To address the centralized nature of most smart city deployments, distributed algorithms should be explored but existing models are unable to explore satisfactory ideas for energy efficiency⁴⁷.

Future research direction

The goal of energy optimization clustering IoT is to solve several issues with conventional IoT systems, including fault tolerance, scalability, energy efficiency, and adaptability. It helps create IoT systems that are capable of self-organization, self-healing, and environment adaptation by taking cues from blockchain systems.

1. Hybrid approaches: The abilities of IoT can be improved by combining algorithms with additional computing methods like machine learning, data analytics, and deep learning.
2. Integration with edge computing: Some algorithmic programs may be utilized at the sharpness to allow thinking deciding and optimization of resources, particularly arising emphasis on some computation to IoT.
3. Collaborative IoT systems: Collaborative intelligent management and optimization inspired by biology. A broad investigation that closely blends theory and practice is part of it.

IoT Techniques	Role of the techniques	Limitation	Matrix formation	Talks about IoT	Talks about Blockchain	Talks about Blockchain formulas	Talks about IoT inspired by blockchain	The Prospects for Blockchain	Future directions and challenges for Blockchain-based IoT are explored
13	To minimize the energy expenditure and maximize the lifetime of IoT nodes	Cost-effective	○	●	●	○	○	●	○
24	Recovery of rechargeable battery degradation job	Need more energy	●	●	○	●	○	○	○
25	Implementation of Energy Awareness	Cost-effective	●	●	●	●	○	○	○
26	To extend its lifetime and lower its energy usage	Need more energy	●	●	○	●	○	○	○
27	To address LEACH shortcomings	Cost-effective	●	●	●	●	○	○	○
28	Minimize the energy expenditure	Lack of privacy	●	○	●	○	○	○	○
29	To solve the hot spot problem in IoT	Cost-effective	○	●	○	●	○	●	●
30	Maintaining the connectivity between the clusters and extending the lifespan of CHs nearer the base station	Lack of privacy	●	●	●	●	○	○	○
31	Energy load balancing and the election of CHs	Need more energy	●	●	●	●	○	○	○
32	Apply Energy Efficient intra-clustering in IoT	Cost-effective	●	●	○	●	○	○	○
33	Apply Routing algorithms in IoT	High energy consumption	○	●	●	●	○	○	○

Table 3. Qualitative analysis of various IoT approaches (Yes– ● No– ○).

Reference	Tools	No. of sensors	Accuracy	Outcome
17	Python	10,000	Average	Improved the energy consumption inside the clusters
18	NS3	Cluster form	Average	Decrease energy expenditure
19	NS2	60	Good	The selection of cluster heads was highly accurate
20	Python	100	Average	Decrease energy expenditure
21	NS3 platform	200	Good	Decrease energy expenditure
22	Matlab	12,000	Average	Best (also strongest) peer cluster distribution
23	Python	800	Good	Find latency and energy consumption

Table 4. Comparison of simulation analysis with various blockchain-based IoT approaches.

4. Explainability and interpretability: The need for understandable and interpretable models will grow as bio-inspired IoT grows more sophisticated and self-sufficient.

Insight thought from this article

Blockchain-based strategies for IoT in energy consumption management present some advantages as well as difficulties. Firstly, blockchain technology facilitates decentralized control and offers a transparent ledger for monitoring the consumption of energy and transactions in IoT networks. By being transparent, participants are more likely to trust one another and there is less chance of fraud or manipulation. Smart contracts facilitate the automation of energy transactions by empowering devices to trade energy on their own, subject to predetermined parameters.

Blockchain enables IoT devices to purchase and sell surplus electricity in real-time on a peer-to-peer basis. As a result, the energy grid becomes more adaptable and durable, encouraging the integration of renewable energy sources and lowering reliance on centralized utilities. Blockchain ensures the integrity and privacy of energy data shared amongst IoT devices with strong security features like hashing using cryptography and consensus mechanisms. Finding a balance between data transparency requirements and privacy concerns is still difficult, though. As the number of IoT gadgets and transactions rises, scalability is still a problem for blockchain-based energy systems. Off-chain protocols and splitting are two examples of solutions being investigated to improve scalability without sacrificing security or decentralization. As blockchain-based energy systems become more prevalent, governments are trying to strike a balance between innovation, consumer protection, and grid stability. To manage possible risks and compliance concerns and promote investment and adoption, clear laws and regulations are essential.

Conclusion

In IoT, blockchain has been used as a popular topology management strategy. Although clustering is primarily used as a way to reduce energy consumption. Numerous clustering strategies address different networking issues such as load balancing, quality of service, security, and mobility management. There is a need to use a hybrid

model for energy efficiency in blockchain-based IoT techniques. It is also concluded that integrating IoT devices with blockchain technology offers several chances for optimizing energy and encouraging renewable energy adoption with some key challenges like scalability, interoperability, and regulatory barriers. To overcome these challenges and fully realize the potential of blockchain technology in IoT energy management, cooperation amongst industry players, legislators, and developers of technology will be essential. It also points out areas for future study and issues that must be resolved to fully realize the promise of blockchain-based solutions for influencing energy systems in the future.

Data availability

All data generated or analyzed during this study are included in this published article.

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Author contributions

Sk Md Habibullah, Sahabul Alam, and Arindam Dey wrote the main manuscript text. Sk Md Habibullah, Sahabul Alam, Arindam Dey, Shivnath Ghosh, and Anurag De prepared all figures and literature survey. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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